

NHDF Hybrid Validation Record

Complete Qwen3-30B-A3B inference on a 12 GB RTX 5070 Ti Laptop GPU

Final disposition: VALIDATED
The complete 30,532,122,624-parameter model produced correct, coherent outputs through the normal fail-closed NHDF launcher. The externally authored IQ2_M payload peaked at 10,487 MiB on the 12,227 MiB GPU and measured 102.37 generated tokens/s in the final 64-token llama-bench run.

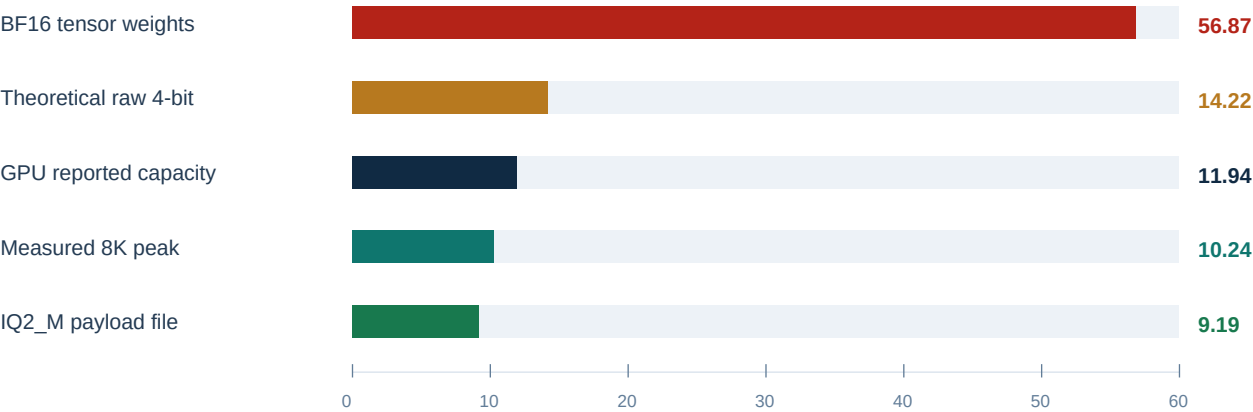
Document field	Value
Artifact	qwen3-30b-a3b-nhdf-v03-iq2m
Artifact format	nhdf-edge-hybrid-0.1
Validation scope	Functional smoke, 49/49 GPU offload, allocated-8K residency, resource margin, and 64-token throughput
Created UTC	2026-08-31T11:59:48.850211Z
Gate UTC	2026-08-31T12:01:34.628224Z
Target	NVIDIA GeForce RTX 5070 Ti Laptop GPU
Manifest SHA-256	669f31e152d74211d6f983fe66c7ea8a7a6dceb7f9f1bc426771cb8aeafdded5
Evidence SHA-256	ab88bcaa91a7fd22f84ec669579356c216fc4ab7ce1f654f4f5cc4fdb0458316

Authoritative evidence lives in the JSON manifest and functional gate record. This PDF is a human-readable technical record generated from those files.

1. Outcome and literal accessibility benefit

The target was not merely to open a model file. Success required the complete model to produce useful language, remain inside the laptop GPU resource envelope, and meet a declared generation-speed floor through an NHDF-controlled launch path. That target was achieved using an external mixed-bit codec inside an NHDF v0.3-informed substrate artifact.

Storage and device capacity (GiB)



Measurement	Final value	Interpretation
Model parameters	30,532,122,624	Complete Qwen3-30B-A3B state; about 3.3B parameters are active per token.
BF16 tensor bytes	61,064,245,248 (56.87 GiB)	Weights alone are 4.76x the reported GPU capacity; practical BF16 inference requires a 64 GiB-class device.
IQ2_M payload file	9,870,270,464 (9.19 GiB)	6.1867x smaller; 83.84% reduction.
Bytes removed	51,193,974,784	47.68 GiB fewer serialized weight bytes than BF16 tensor data.
Measured allocated-8K peak	10,487 MiB	Leaves 1,740 MiB of the reported 12,227 MiB.
Generation microbenchmark	102.37 +/- 3.59 tok/s	Three 64-token llama-bench samples; exceeds the fixed 80 tok/s floor.

Accessibility in plain terms

A workload whose BF16 weights alone need 56.87 GiB now runs on this laptop's 11.94 GiB reported device capacity. Even an ideal raw 4-bit representation would be 14.22 GiB before metadata and runtime buffers, so a conventional 4-bit floor would still miss the target.

2. What NHDF did - and did not do

Responsibility boundary



Compression belongs to Bartowski/ggml IQ2_M. NHDF supplies policy, binding, evidence, and fail-closed launch.

The weight encoding is GGUF/IQ2_M from Bartowski and ggml. It is executed by llama.cpp b6014. NHDF does not claim authorship of that codec. In this result, NHDF is the controlling substrate: it binds exact artifacts and runtime files, records typed validation state, enforces GPU/context policy, stores sealed evidence, verifies its event chain, and refuses ordinary execution until promotion evidence passes.

Responsibility	Owner	Concrete implementation
Low-bit weight encoding	Bartowski / ggml	GGUF IQ2_M mixed-bit payload, explicitly attributed in the manifest.
Tensor execution	llama.cpp	Pinned b6014 CUDA build for compute capability 12.0.
Artifact binding	NHDF hybrid	Path, byte count, SHA-256, source revision, runtime file hashes, and specification hash.
Resource policy	NHDF hybrid	Exact GPU identity, context <= 8192, q8 K/V, full offload, 10,712 MiB free-memory preflight, and 512 MiB reserve.
Promotion	NHDF hybrid	UNCALIBRATED until functional, residency, offload, resource, and throughput gates all pass.
Deployment launch	NHDF hybrid	Fails closed for altered, missing, unvalidated, wrong-GPU, or under-resourced artifacts.

Honest boundary
This is a successful NHDF-controlled hybrid execution profile. It is not proof of an NHDF-native weight codec, broad benchmark accuracy, or quality after filling all 8K context positions.

3. Fresh validation procedure and outputs

3.1 Promotion sequence

The final artifact was recreated in UNCALIBRATED state. A normal run was confirmed to refuse launch. The gate then performed a full payload hash check, five cold model launches, four 512-token functional prompts, one allocated-8192 cache run with a short exact-response prompt, and three prompt plus three generation benchmark samples. Only the complete pass wrote VALIDATED.

Gate	Required	Observed	Result
exact_ok	OK	OK	PASS
arithmetic	^323\\.?\$	323	PASS
integrity_vs_quality	required terms	A checksum passing only verifies data integrity, not the model's ability to generate meaningful or accurate outputs, as corruption could still exist in ways that don't affect the checksum but degrade performance.	PASS
code	required terms	```python def is_even(n): return n % 2 == 0 ```	PASS

3.2 Residency and buffer observations

Resource	Measured value
Physical GPU capacity	12,227 MiB
Idle baseline in canonical run	287 MiB
Peak system-wide device use	10,487 MiB
Measured headroom	1,740 MiB
CUDA model buffer	9279.83 MiB
CUDA q8 K/V buffer	408.00 MiB
CUDA compute buffer	300.75 MiB
CPU-mapped model buffer	127.51 MiB
Reported offload	49/49 layers

The 8K gate allocates an 8192-position q8 K/V cache and executes a short prompt. It proves allocation, residency, offload, and short execution at that capacity. It does not measure filled-8K retrieval or answer quality.

4. Throughput and operational behavior

Workload	Samples (tok/s)	Mean	Std. dev.	Gate
64-token prompt	457.008, 468.031, 450.537	458.525658	8.845115	Recorded
64-token generation	98.2264, 104.2680, 104.6090	102.367894	3.590717	>= 80 PASS

Interpretation of 102.37 tok/s

This is a llama-bench batch-one, 64-token generation microbenchmark on a mixture-of-experts model. All 30.532B parameters are stored, but about 3.3B are active per token. The result is therefore not dense-30B throughput and is not an 'optimal under every workload' claim.

4.1 Normal validated launch

After promotion and full byte-for-byte verification, the ordinary launcher was run without --allow-unvalidated. It returned:

The validated NHDF hybrid enables efficient, high-performance neural network inference directly on this laptop's GPU.

4.2 Fail-closed resource behavior observed during development

One intermediate retry detected a transient system-wide GPU allocation during the 8K launch: peak use reached 12,088 MiB and only 139 MiB remained. Functional prompts and throughput still passed, but promotion was rejected because the 512 MiB reserve failed. Idle use then returned to 287 MiB; the artifact was recreated and the final canonical gate passed at 10,487 MiB. This observation is not the canonical certificate, but it demonstrates why system-wide peak sampling and fail-closed resource disposition matter.

5. What failed before the successful hybrid

5.1 Legacy NHDF-native scalar pack - functional failure

The original 9,152,386,624-byte native tensor pack loaded, passed 531 manifest/CRC/parity checks, fit the GPU, and executed CUDA generation. Its output collapsed to repeated newline tokens or the string 10000000. It is sealed as QUALITY_FAILED and the native loader refuses it by default. Integrity and fit were real; language quality failed.

Layer-0 expert	Output NRMSE	Cosine similarity
0	0.4813	0.8805
17	0.4749	0.8935
127	0.4482	0.9170

5.2 Replacement native codec - comparative failure, not collapse

A later routed-calibrated 3-bit GPTQ experiment achieved useful numerical behavior. On the complete router-weighted layer-0 gate over 656 disjoint holdout tokens and all 117 experts selected by real top-8 routes, its NRMSE was 0.155101 versus 0.150849 for simpler equal-storage RTN. The candidate was 2.819% worse. Absolute quality passed, but the predeclared comparative-advantage gate failed, so a native full pack was not justified.

Correct conclusion

The native scalar format genuinely failed language generation. The replacement native candidate did not collapse, but it failed to beat a simpler equal-budget codec. Neither negative result invalidates the separately validated external-codec hybrid.

6. Reproduce the validated path

6.1 Restore large external inputs

The Git repository deliberately excludes the 9.87 GB GGUF, 61.07 GB BF16 source, build trees, temporary calibration data, and profiler captures. The exact validated llama.cpp binaries, their license, and their provenance record are included because the sealed runtime set remains below the repository publication budget. Restore only the IQ2_M model file at the path below and verify both byte count and SHA-256 before creating the hybrid.

Input	Pinned value
Model repository	bartowski/Qwen_Qwen3-30B-A3B-Instruct-2507-GGUF
Model revision	6c6e8692f43e4ca663f7ece8229a1361090d3a4c
Model artifact	Qwen_Qwen3-30B-A3B-Instruct-2507-IQ2_M.gguf
Model bytes	9,870,270,464
Model SHA-256	f2dc78edd3ec0171904f1945d8c05a948131b1103172b1710b763db2eb65f52a
llama.cpp source	https://codeload.github.com/ggml-org/llama.cpp/zip/refs/tags/b6014
llama.cpp source archive SHA-256	936b96528b12d06928e9c2e8f929dfae4eb25923faa46f49de23f811fa35f92a
Build profile	Ninja; CUDA 12.8.61; SM 120; MSVC 19.44.35221.0

Use scripts/download_verified_ranges.ps1 with the immutable_url, byte count, and hash in CONTROL_SOURCE.json when segmented downloading is useful.

6.2 Create, gate, verify, and run

```
nhdf-edge create-hybrid packs\qwen3-30b-a3b-nhdf-v03-iq2m `
  --model models\Qwen3-30B-A3B-Instruct-2507-IQ2_M\Qwen_Qwen3-30B-A3B-Instruct-2507-IQ2_M.gguf `
  --runtime tools\llama.cpp-b6014\bin\llama-cli.exe `
  --benchmark-runtime tools\llama.cpp-b6014\bin\llama-bench.exe `
  --specification sources\NHDF_Formal_Specification_v0.3_General_Purpose_CCD_Tom_Klootwijk.pdf `
  --source-record models\Qwen3-30B-A3B-Instruct-2507-IQ2_M\CONTROL_SOURCE.json `
  --assurance-evidence metrics\local\gguf_backend_ops.json

nhdf-edge gate-hybrid packs\qwen3-30b-a3b-nhdf-v03-iq2m
nhdf-edge verify packs\qwen3-30b-a3b-nhdf-v03-iq2m
nhdf-edge run packs\qwen3-30b-a3b-nhdf-v03-iq2m --quick --text-only `
  --prompt "Reply with exactly the single word OK." --max-new-tokens 8
```

A new artifact begins UNCALIBRATED. Do not use --allow-unvalidated for deployment. The final gate must be rerun on the target machine because VRAM headroom and throughput are hardware- and background-load-dependent.

Appendix A. Identity and integrity records

Record	Value
Upstream Qwen revision	0d7cf23991f47feeb3a57ecb4c9cee8ea4a17bfe
GGUF revision	6c6e8692f43e4ca663f7ece8229a1361090d3a4c
GGUF SHA-256	f2dc78edd3ec0171904f1945d8c05a948131b1103172b1710b763db2eb65f52a
NHDF v0.3 specification SHA-256	019f7da2b97cd73b169e8133b52f4d4bd4d687f9444281364fdb962977ab0956
Manifest SHA-256	669f31e152d74211d6f983fe66c7ea8a7a6dceb7f9f1bc426771cb8aeafdded5
Functional evidence SHA-256	ab88bcaa91a7fd22f84ec669579356c216fc4ab7ce1f654f4f5cc4fdb0458316
Event-chain records verified	2
Backend assurance	8,132 / 8,132 CUDA MUL_MAT_ID cases passed; separately sealed assurance evidence

Sealed runtime file	Bytes	SHA-256
llama-cli.exe	1,228,800	ceef9666f97d8fc260557217560fd1c3e62ffdc2c2b7acb457b8c8d21f1508ac
llama-bench.exe	219,648	49a935fd8f150810b28507638c7731c4f872346f5b2ecbacc1d2dfc4362edebc
ggml-base.dll	510,976	254eceb789874fac44f03cdd7894b3fb5a9d94cd1710528861b3c1ca4c5b21df
ggml-cpu.dll	630,272	0f1dbc3cdc7fdd2d27bc5bf1ce1a84321fa2044a2b2110614c96be5f01f20f7a
ggml-cuda.dll	35,027,456	5c69804a4ab1ee606e4ab903a5148b38753f9e03eec52125539ef0384268f82f
ggml.dll	67,584	90ba4c079d3d423a534a523c8f0f983f2d6443ca23f682108916a9d017f1988a
llama.dll	1,551,872	0f8ed0e4819dd5d8e1700cc85add08b10c3d35ecd2f36acbed95d3b87f00ea1d

These are SHA-256-sealed local records, not externally signed attestations. Recomputing both a file and its local digest is possible for a repository writer; use signed Git commits or an external transparency log if adversarial provenance becomes a requirement.

Appendix B. Evidence boundary and next work

Established on this machine

- Complete-model loading and coherent generation through the NHDF launcher.
- Exact payload, runtime, specification, source-record, and evidence hashes.
- Four of four declared functional prompts passed.
- Forty-nine of forty-nine layers offloaded in every canonical launch.
- Allocated-8K q8 K/V residency at 10,487 MiB peak with 1,740 MiB headroom.
- Final 64-token generation microbenchmark at 102.367894 +/- 3.590717 tok/s.
- Fail-closed status, GPU identity, free-memory, context, reserve, and throughput policy.
- Sixty-two Python tests passed after the final implementation changes.

Not established

- Broad benchmark accuracy, perplexity, or KL agreement against BF16.
- Retrieval or language quality after actually filling 8K context positions.
- Sustained thermals, power, p10/p90 performance, or long-duration stability.
- Dense-30B throughput equivalence.
- An NHDF-native tensor-codec advantage.
- A self-contained portable artifact; the validated zero-copy manifest is workspace-bound.
- Performance portability to other GPUs or llama.cpp revisions.

Recommended next gates

Priority	Next gate	Why
1	Broad quality suite against BF16 or a stronger reference	Turns the narrow smoke certificate into an evidence-backed quality statement.
2	Filled-context tests at 2K, 4K, and 8K	Separates allocated capacity from usable long-context quality and throughput.
3	Sustained 10-30 minute thermal and power runs	Quantifies laptop throttling and stable tokens/s rather than cold microbenchmarks.
4	Signed release manifest or external transparency record	Strengthens provenance beyond local digest files.
5	Native-codec calibration-only selector versus RTN	Continue native research only where a candidate beats the simpler equal-budget baseline.

Bottom line

The practical objective was achieved: the complete 30.532B Qwen model is accessible and useful on the tested 12 GB laptop GPU through an NHDF-controlled, externally encoded hybrid. The exact success, failures, hashes, commands, measurements, and limitations are preserved here for the next iteration.